MENELAT 30 MG TABLETS
MENELAT 45 MG TABLETS

(Mirtazapine Tablets U.S.P. 30 mg & 45 mg)

BRAND OR PRODUCT NAME
MENELAT 30 MG TABLETS
MENELAT 45 MG TABLETS

NAME AND STRENGTH OF ACTIVE SUBSTANCE(S)

MENELAT 30 MG TABLETS
Each film coated tablet contains:
Mirtazapine Ph.Eur. 30 mg

MENELAT 45 MG TABLETS
Each film coated tablet contains:
Mirtazapine Ph.Eur. 45 mg

PRODUCT DESCRIPTION

MENELAT 30 MG TABLETS
Pink colored, round, biconvex film coated tablets plain on one side and break line on other side.

MENELAT 45 MG TABLETS
White to off white colored, round, biconvex film coated tablets plain on both sides.

DOSAGE FORM
Film coated tablets

CLINICAL PHARMACOLOGY

PHARMACODYNAMIC
Pharmacotherapeutic group: Other antidepressants, ATC code: N06AXl l
Mirtazapine is a centrally active presynaptic α 2-antagonist, which increases central noradrenergic and serotonergic neurotransmission. The enhancement of serotonergic neurotransmission is specifically mediated via 5-HT1 receptors, because 5-HT2 and 5-HT3 receptors are blocked by mirtazapine. Both enantiomers of mirtazapine are presumed to 373 18A contribute to the antidepressant activity, the S (+) enantiomer by blocking α 2 and 5-HT2 receptors and the R (-) enantiomer by blocking 5-HT3 receptors.
The histamine H1-antagonistic activity of mirtazapine is associated with its sedative properties. It has practically no anticholinergic activity and, at therapeutic doses, has practically no effect on the cardiovascular system.

PHARMACOKINETIC
After oral administration of Mirtazapine, the active substance mirtazapine is rapidly and well absorbed (bioavailability - 50 %), reaching peak plasma levels after approx. two hours. Binding of mirtazapine to plasma proteins is approx. 85 %. The mean half-life of elimination is 20·40 hours; longer half-lives, up to 65 hours, have occasionally been recorded and shorter half-lives have been seen in young men. The half-life of elimination is sufficient to justify once-a-day dosing. Steady state is reached after 3-4 days, after which there is no further accumulation. Mirtazapine displays linear pharmacokinetics within the recommended dose range. Food intake has no influence on the pharmacokinetics of mirtazapine. Mirtazapine is extensively metabolized and eliminated via the urine and feces within a few days. Major pathways of biotransformation are demethylation and oxidation, followed by conjugation. In vitro data from human liver microsomes indicate that Cytochrome P450 enzymes CYP2D6 and CYP1A2 are involved in the formation of the 8-hydroxy metabolite of mirtazapine, whereas CYP3A4 is considered to be responsible for the formation of the N-demethyl and N-oxide metabolites. The demethyl metabolite is pharmacologically active and appears to have the same pharmacokinetic profile as the parent compound.
The clearance of mirtazapine may be decreased as a result of renal or hepatic impairment.

INDICATION
Episode of major depression

CONTRAINDICATION
Hypersensitivity to the active substance or to any of the excipients.
Concomitant use of mirtazapine with monoamine oxidase (MAO) inhibitors.

WARNINGS AND PRECAUTION

Use in children and adolescents under 18 years of age
Mirtazapine should not be used in the treatment of children and adolescents under the age of 18 years. Suicide-related behavior (suicide attempt and suicidal thoughts), and hostility (predominantly aggression, oppositional behavior and anger) were more frequently reported. If, based on clinical need, a decision to treat is nevertheless taken; the patient should be carefully monitored for the appearance of suicidal symptoms. In addition, long-term safety data in children and adolescents concerning growth, maturation and cognitive and behavioral development are lacking.

Suicide / suicidal thoughts or clinical worsening
Depression is associated with an increased risk of suicidal thoughts, self harm and suicide (suicide-related events). This risk persists until significant remission occurs. As improvement may not occur during the first few weeks or more of treatment, patients should be closely monitored until such improvement occurs. It is general clinical experience that the risk of suicide may increase in the early stages of recovery. Patients with a history of suicide-related events or those exhibiting a significant degree of suicidal ideation prior to commencement of treatment are known to be at greater risk of suicidal thoughts or suicide attempts, and should receive careful monitoring during treatment. Close supervision of patients and in particular those at high risk should accompany therapy with antidepressants especially in early treatment and following dose changes. Patients (and caregivers of patients) should be alerted about the need to monitor for any clinical worsening, suicidal behavior or thoughts and unusual changes in behavior and to seek medical advice immediately if these symptoms present. With regard to the chance of suicide, in particular at the beginning of treatment, only a limited number of Mirtazapine tablets should be given to the patient.

Bone marrow depression
Bone morrow depression, usually presenting as granulocytopenia or agranulocytosis, has been reported during treatment with Mirtazapine. In the postmarketing period with Mirtazapine very rare cases of agranulocytosis have been reported, mostly reversible, but in some cases fatal. Fatal cases mostly concerned patients with on age above 65. The physician should be alert for symptoms like fever, sore throat, stomatitis or other signs of infection; when such symptoms occur, treatment should be stopped and blood counts taken.

Jaundice
Treatment should be discontinued if jaundice occurs.

Conditions which need supervision
Careful dosing as well as regular and close monitoring is necessary in patients with:
- Epilepsy and organic brain syndrome: Menelat should be introduced cautiously in patients who have a history of seizures. Treatment should be discontinued in any patient who develops seizures, or where there is an increase in seizure frequency.
- Hepatic impairment: Following a single 15 mg oral dose of mirtazapine, the clearance of mirtazapine was decreased in mild to moderate hepatically impaired patients and the average plasma concentration of mirtazapine was increased.
- Renal impairment: Following a single 15 mg oral dose of mirtazapine, in patients with and severe renal impairment the clearance of mirtazapine was about decreased.
- Cardiac diseases like conduction disturbances, angina pectoris and recent myocardial infarction, where normal precautions should be taken and concomitant medicines carefully administered.
- Low blood pressure.
- Diabetes mellitus: In patients with diabetes, antidepressants may alter glycemic control. Insulin and/or oral hypoglycemic dosage may need to be adjusted and close monitoring is recommended.

Like with other antidepressants, the following should be taken into account.
- Worsening of psychotic symptoms can occur when antidepressants are administered to patients with schizophrenia or other psychotic disturbances: paranoid thoughts can be intensified.
- When the depressive phase of bipolar disorder is being treated, it can transform into the manic phase. Patients with a history of mania/hypomania should be closely monitored. Mirtazapine should be discontinued in any patient entering a manic phase.
- Although Mirtazapine is not addictive, post-marketing experience report that abrupt termination of treatment after long term administration may sometimes result in withdrawal symptoms. The majority of withdrawal reactions are mild and self-limiting. Among the various reported withdrawal symptoms dizziness, agitation, anxiety, headache and nausea are the most frequently reported. Even though they have been reported as withdrawal symptoms, it should be realized that these symptoms may be related to the underlying disease. It is recommended to discontinue treatment with mirtazapine gradually.
- Care should be taken in patients with micturition disturbances like prostate hypertrophy and in patients with acute narrow angle glaucoma and increased intra-ocular pressure (although there is little chance of problems with Mirtazapine because of its very weak anticholinergic activity).
- *Akathisia/psychomotor restlessness:* The use of antidepressants has been associated with the development of akathisia, characterized by a subjectively unpleasant or distressing restlessness and need to move often accompanied by on inability to sit or stand still. This is most likely to occur within the first few weeks of treatment. In patients who develop these symptoms, increasing the dose may be detrimental.

Hyponatremia
Hyponatremia has been reported very rarely with the use of mirtazapine. Caution should be exercised in patients at risk, such as elderly patients or patients concomitantly treated with medications known to cause hyponatremia.

Serotonin syndrome
Interaction with serotonergic active substances: Serotonin syndrome may occur when selective serotonin reuptake inhibitors (SSRIs) are used concomitantly with other serotonergic active substances. Symptoms of serotonin syndrome may be hyperthermia, rigidity, myoclonus, autonomic instability with possible rapid fluctuations of vital signs, mental status changes that include confusion, irritability and extreme agitation progressing to delirium and coma. From post marketing experience it appears that serotonin syndrome occurs very rarely in patients treated with Mirtazapine alone.

Elderly patients
Elderly patients are often more sensitive, especially with regard to the undesirable effects of antidepressants. During clinical research with Mirtazapine, undesirable effects have not been reported more often in elderly patients than in other age groups.

DRUG INTERACTION

Pharmacodynamic Interactions
- Mirtazapine should not be administered concomitantly with MAO inhibitors or within two weeks after discontinuation of MAO inhibitor therapy. In the opposite way about two weeks should pass before patients treated with mirtazapine should be treated with MAO inhibitors. In addition, as with, co-administration with other serotonergic active substances (L-tryptophan, triptans, tramadol, linezolid, SSRIs, Venlafaxine, lithium and St. John's Wort - Hypericum perforatum - preparations) may lead to an incidence of serotonin associated effects. Caution should be advised and a closer clinical monitoring is required when these active substances are combined with mirtazapine.
- Mirtazapine may increase the sedating properties of benzodiazepines and other sedatives (notably most anti psychotics, antihistamine HI antagonists, opioids). Caution should be exercised when these medicinal products are prescribed together with mirtazapine.
- Mirtazapine may increase the CNS depressant effect of alcohol. Patients should therefore be advised to avoid alcoholic beverages while taking mirtazapine.

Pharmacokinetic interactions
- Carbamazepine and phenytoin, CYP3A4 inducers, increased mirtazapine clearance about twofold, resulting in a decrease in average plasma mirtazapine concentration of 60 % and 45 %, respectively. When carbamazepine or any other inducer of hepatic metabolism (such as rifampicin) is added to mirtazapine therapy, the mirtazapine dose may have to be increased. If treatment with such medicinal product is discontinued, it may be necessary to reduce the mirtazapine dose.
- Co-administration of the potent CYP3A4 inhibitor ketoconazole increased the peak plasma levels and the AUC of mirtazapine by approximately 40% and 50% respectively.
- When cimetidine (weak inhibitor of CYP 1A2, CYP2D6 and CYP3A4) is administered with mirtazapine, the mean plasma concentration of mirtazapine may increase more than 50 %. Caution should be exercised and the dose may have to be decreased when co-administering mirtazapine with potent CYP3A4 inhibitors, HIV protease inhibitors, azole antifungals, erythromycin, cimetidine or nefazodone.
- Interaction studies did not report any relevant pharmacokinetic effects on concurrent treatment of mirtazapine with paroxetine, amitriptyline, risperidone or lithium.

ADVERSE EFFECTS
Depressed patients display a number of symptoms that are associated with the illness itself. It is therefore sometimes difficult to ascertain which symptoms are a result of the illness itself and which are a result of treatment with Mirtazapine.
The most commonly reported adverse reactions are somnolence, sedation, dry mouth, weight increased, and increase in appetite, dizziness and fatigue.
Table1 shows the categorized incidence of the adverse reactions added with adverse reactions from spontaneous reporting.

System organ class	Very common	Common	Uncommon	Rare	Frequency not known
Blood and the lymphatic system disorder					• Bone marrow depression (granulocytopenia, agranulocytosis, aplastic anemia thrombocytopenia) Eosinophilia.
Endocrine disorders					Inappropriate antidiuretic hormone secretion
Metabolism and nutrition disorders	• Weight increased • Increase in appetite				• Hyponatraemia
Psychiatric disorder		• Abnormal dreams Confusion Anxiety Insomnia	• Nightmares • Mania • Agitation • Hallucinations • Psychomotor restlessness (incl. akathisia, hyperkinesias)	aggression	• Suicidal ideation • Suicidal behavior
Nervous system disorder	• Somnolence • Sedation • Headache	• Lethargy • Dizziness • Tremor • Amnesia	• Paraesthesia • Restless legs • Syncope	• Myoclonus	• Convulsions (insults) • Serotonin syndrome • Oral paraesthesia dysarthria
Vascular disorder		• Orthostatic hypotension	• Hypotension		
Gastrointestinal disorder	• Dry mouth	• Nausea • Diarrhea • Vomiting	• Oral hypoasthesia	pancreatitis	• Mouth oedema, Increased salivation
Hepatobiliary disorders				• Elevations in serum transaminas e activities	
Skin and subcutaneous tissue disorders		• Exanthema			• Stevens-Johnson Syndrome, dermatitis bullous, Erythema multiforme, Toxic epidermal necrolysis. • Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS).
Musculoskeletal and connective tissue disorders and bone disorders		• Arthralgia • Myalgia • Back pain			
General disorders and administration site conditions		• Oedema peripheral • Fatigue			Somnambulism

In laboratory evaluations transient increases in transaminases and gamma-glutamyltransferase have been reported.

OVERDOSE
Present experience concerning overdose with Mirtazapine alone indicates that symptoms are usually mild. Depression of the central nervous system with disorientation and prolonged sedation has been reported, together with tachycardia and mild hyper or hypotension. However, there is a possibility of more serious outcomes (including fatalities) at dosages much higher than the therapeutic dose, especially with mixed overdoses.
Cases of overdose should receive appropriate symptomatic and supportive therapy for vital functions. Activated charcoal or gastric lavage should also be considered.

DOSAGE AND ADMINISTRATION

Adults
Treatment should begin with 15 mg daily. The dosage generally needs to be increased to obtain an optimal clinical response. The effective daily dose is usually between 15 and 45 mg (the dose should be taken at night).
Mirtazapine begins to exert its effect in general after 1-2 weeks of treatment. Treatment with an adequate dose should result in a positive response within 2-4 weeks. With an insufficient response, the dose can be increased up to the maximum dose. If there is no response within a further 2-4 weeks, then treatment should be stopped.

Elderly
The recommended dose is the same as that for adults. In elderly patients an increase in dosing should be done under close supervision to elicit a satisfactory and safe response.

Children and adolescents under the age of 18 years
Mirtazapine should not be used in children and adolescents under the age of 18 years.

Renal impairment
The clearance of mirtazapine may be decreased in patients with moderate to severe renal impairment (creatinine clearance < 40 ml /min). This should be taken into account when prescribing Mirtazapine to this category of patients.

Hepatic impairment
The clearance of mirtazapine may be decreased in patients with hepatic impairment. This should be taken into account when prescribing Mirtazapine to this category of patients, particularly with severe hepatic impairment. As patients with severe hepatic impairment have not been investigated.
Mirtazapine has an elimination half-life of 20-40 hours and therefore Mirtazapine is suitable for once daily administration. It should be taken preferably as a single night-time dose before going to bed. Mirtazapine may also be given in two divided doses (once the morning and once at night-time, the higher dose should be taken at night).
The tablets should be taken orally. The tablet will and can be swallowed without water.
Patients with depression should be treated for a sufficient period of at least 6 months to ensure that they are free from symptoms.
It is recommended to discontinue treatment with mirtazapine gradually to avoid withdrawal symptoms.

INTERFERENCE WITH COGNITIVE AND MOTOR PERFORMANCE
Mirtazapine may impair judgement, thinking, and particularly, motor skills, because of its prominent sedative effect. The drowsiness associated with mirtazapine use may impair a patient's ability to drive, use machines or perform tasks that require alertness. Thus, patients should be cautioned about engaging in hazardous activities until they are reasonably certain that Mirtazapine therapy does not adversely affect their ability to engage in such activities.

PREGNANCY AND LACTATION
Limited data of the use of mirtazapine in pregnant women do not indicate an increased risk for congenital malformations. Caution should be exercised when prescribing to pregnant women. If Mirtazapine is used until, or shortly before birth, postnatal monitoring of the newborn is recommended to account for possible discontinuation effects. A decision on whether to continue/discontinue breast-feeding or to continue/discontinue therapy with Mirtazapine should be made taking into account the benefit of breast-feeding to the child and the benefit of Mirtazapine therapy to the woman.

STORAGE CONDITIONS
Do not store above 30°C. Protect from light and moisture.

PACKAGING AVAILABLE
Menelat 30mg & 45mg Tablets are packed in Alu-Alu blister of 10 tablets. Such blisters containing 10 Tablets are packed into boxes of 1x10 Tablets, 3x10 Tablets and 10x10 Tablets.
"Not all packs will be available commercially."

DATE OF REVISION OF PACKAGE INSERT
29.09.2021

torrent PHARMA

Manufactured by :
TORRENT PHARMACEUTICALS LTD.
Indrad-382 721, Dist. Mehsana, INDIA.

PRODUCT NAME :	Menelat	**COUNTRY :** Malaysia	**LOCATION :** Indrad	**Supersedes A/W No.:**			
ITEM / PACK :	Insert	**NO. OF COLORS:** 1	**REMARK : Folded Size 35 mm**				
DESIGN STYLE :	Front Back	**PANTONE SHADE NOS.:**	**SUBSTRATE :**				
CODE :	xxxxxxxxx-5253	Black	**Activities**	**Department**	**Name**	**Signature**	**Date**
DIMENSIONS (MM) (LxWxH) :	180 x 330 mm		Prepared By	**Pkg.Dev**			
ART WORK SIZE :	S/S		Reviewed By	**Pkg.Dev**			
DATE :	29-09-2021		Approved By	**Quality**			

This colour proof is not colour binding. Follow Pantone shade reference for actual colour matching.